DD_QA-01 - End-to-End Test Strategy

Version: v1.0

Status: Draft for review

Bundle: `00_index`

Depends on:

- `DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0`
- `DD_API-00-API_Catalog_and_Gateway_Standards-v1.0`
- All backend module DDs participating in tested journeys
- All frontend DDs participating in tested journeys

1. Purpose

This DD defines the minimum end-to-end testing strategy for SOPHIX V3.

The goal is to prove that modules work together as a BSS, not only that each module works alone.

2. Test Layers

Layer	Purpose	Owner
Unit tests	Validate local service/rule behavior.	Module squad.
API contract tests	Verify OpenAPI request/response compatibility.	Module squad + frontend consumers.
Event contract tests	Verify produced/consumed Kafka payloads.	Producer and consumer squads.
Integration tests	Verify module with DB/cache/Kafka/Camunda/Drools.	Module squad.
Cross-module E2E tests	Verify business journey across modules.	QA/platform with module squads.
Frontend smoke tests	Verify role-based screens and API wiring.	Frontend squad + QA.
UAT scenarios	Business validation.	Product/business owners + QA.

Rule: a feature is not complete until its cross-module E2E test passes or the DD explicitly says the feature is module-local.

3. Required Test Data

Create reusable seeded data for WIK first.

Data set	Examples
Operators	`WIK`, optional future `WUG`, `WTZ`, `YASSN` seeds.
Auth users	Customer care, sales agent, sales supervisor, billing operator, dispatcher, contractor tech, catalog admin, platform admin.
Tech regions	`KE-NRB-LAVINGTON`, `KE-NRB-KAREN`, `KE-MOMBASA-NYALI`.
Contractors/teams	Internal team, franchise contractor, field team, sales team.
HomePass	Sellable, not sellable, already occupied, wrong region.
Packages	Active eligible package, inactive package, package not available in target region.
Customers	Clean customer, blocked/fraud customer, existing subscription customer.
Billing	Pending invoice, paid invoice, overdue invoice, wallet balance.
Equipment	Available ONT, assigned ONT, defective equipment, recovered equipment.

4. Minimum E2E Scenarios

4.1 New Customer Sale To Activation

Purpose: proves SALES, FUL, ILM, RLM, SIP, BIL, WO, SUB, OSR, NOT/ICN work together.

Flow:

1. FE-APP-02 creates lead through SALES-01.
2. SALES-01 qualifies lead using RLM/SIP/ILM.
3. FE-APP-02 converts lead to FUL-02 order.
4. FUL-02 creates/links customer/KYC and starts order BPMN.
5. Payment is applied through BIL.
6. KYC is approved through Backoffice user task.
7. Install WO is created and assigned.
8. Contractor finalizes WO with equipment evidence.
9. Subscription is created/activated.
10. Activation succeeds and notification is sent.
11. REP receives events and updates funnel metrics.

Success criteria:

- lead status `CONVERTED`;
- FUL order final state activated/complete;
- subscription active in SUB-LM;
- billing ledger correct;
- WO finalized;
- equipment bound/reconciled;
- customer notification emitted;
- reporting dashboard shows one converted/activated order.

4.2 Duplicate Payment Idempotency

Purpose: proves payment retry is safe.

Flow:

1. Submit payment with same provider reference and same idempotency key twice.
2. Submit payment with same idempotency key but different payload.
3. Verify BIL response and ledger.

Success criteria:

- same request returns original result;
- different request hash returns `409`;
- wallet/invoice balance credited once.

4.3 Ticket To Support Work Order

Purpose: proves TCK and WO integration.

Flow:

1. Customer care creates ticket in TCK.
2. Ticket category allows field support.
3. Backoffice creates WO from ticket.
4. WO is assigned and finalized.
5. TCK consumes WO finalization event and updates timeline/status.

Success criteria:

- TCK ticket exists and is linked to WO;
- WO exists and has originating ticket context;
- ticket timeline includes WO created/finalized events;
- ticket can be resolved after WO completion.

4.4 Dunning To Suspension

Purpose: proves billing-to-subscription/fulfillment control.

Flow:

1. Create overdue invoice state.
2. BIL-04 advances dunning level.
3. Suspension threshold reached.
4. BIL-04 starts SUB-WF-SUSPEND-NP.
5. Restriction/suspension fulfillment executes.
6. Customer notification sent.

Success criteria:

- dunning state reaches configured level;
- subscription operation created;
- restriction/suspension state visible;
- notification emitted once;
- reporting dunning metric updated.

4.5 Relocation

Purpose: proves subscription, HomePass, WO, OSR, and billing effects.

Flow:

1. Backoffice starts relocation.
2. SUB-WF validates target HomePass through RLM.
3. WO created for move.
4. Equipment action completed.
5. Subscription location updated after successful workflow.

Success criteria:

- old and new HomePass states are coherent;
- WO finalized;
- subscription references target HomePass;
- equipment state is correct;
- billing effect is applied or explicitly not applicable per DD.

4.6 Equipment Swap/RMA

Purpose: proves WO/OSR equipment lifecycle.

Flow:

1. Support ticket or WO identifies defective equipment.
2. OSR-RMA creates swap request.
3. Contractor installs replacement and recovers old device.
4. Stock movement and instance lifecycle events are recorded.

Success criteria:

- new equipment assigned/bound;
- old equipment recovered/defective/RMA state correct;
- stock balances updated;
- customer/WO view reflects current serial.

4.7 Customer 360 Partial Failure

Purpose: proves frontend composition is resilient.

Flow:

1. Load Customer 360 with all modules healthy.
2. Simulate one read module failing, for example OSR or TCK.
3. Reload Customer 360.

Success criteria:

- working panels render;
- failed panel shows local error/stale state;
- no hidden write action is enabled from missing data;
- correlation id is available.

4.8 Reporting Reconciliation

Purpose: proves REP is accurate enough for management.

Flow:

1. Generate known events for orders, invoices, payments, WOs, tickets.
2. Run REP ingestion and aggregate jobs.
3. Run REP reconciliation jobs against module summary APIs.

Success criteria:

- dashboard totals match source module summaries within documented tolerance;
- mismatches produce reconciliation report;
- export uses file storage and audit.

5. API Contract Testing

Each module must provide tests that validate:

- required headers;
- auth failures;
- role failures;
- validation errors;
- success responses;
- idempotency replay/conflict;
- documented error codes;
- OpenAPI examples.

Frontend squads use generated clients or contract fixtures from OpenAPI where possible.

6. Event Contract Testing

For every event:

Test	Purpose
Schema validation	Producer event matches documented schema.
Idempotent consume	Consumer handles duplicate `eventId`.
Replay	Consumer can reprocess old event safely.
Missing optional field	Consumer uses defaults or rejects with clear error.
Version handling	Consumer accepts compatible schema version.

7. Frontend Smoke Testing

For each frontend app:

App	Required smoke tests
FE-APP-01	Login, role navigation, Customer 360, ticket create, subscription action preview, billing read, WO queue.
FE-APP-02	Login, offline draft, lead create, coverage check, package selection, convert to order.
FE-APP-03	Login, assigned WO list, offline evidence capture, sync, finalize.
FE-APP-04	Login/customer profile, invoice list, payment handoff, ticket create, notification preferences.
FE-APP-05	Dashboard list, dashboard detail, export request, role masking.

8. Test Environments

Environment	Purpose
Local/dev	Module unit/integration tests with local dependencies.
Integration	Cross-module API/event tests with seeded data.
SIT	Full journey testing, near-production topology.
UAT	Business validation with controlled test data.
Performance	Load and concurrency testing for billing, order, WO, reporting, and auth.

9. Entry And Exit Criteria

Entry criteria for cross-module E2E:

- OpenAPI exists for involved APIs;
- test users/roles exist in `FOUNDATION_AUTH`;
- seed data exists;
- Kafka topics/outbox/inbox are configured;
- involved frontend smoke path exists or API-only substitute is approved.

Exit criteria:

- all critical E2E scenarios pass;
- no P1/P2 defects open;
- reconciliation tests pass for billing/subscription/reporting;
- idempotency/retry tests pass for payment/order/subscription commands;
- role/security tests pass.

10. Traceability Matrix

Every E2E test case must reference:

Field	Example
Test id	`E2E-WIK-ACT-001`

Field	Example
Business feature	New customer sale to activation
Frontend DD	`FE-APP-02`, `FE-APP-01`
Backend DDs	SALES, FUL, ILM, RLM, SIP, BIL, WO, SUB, OSR
APIs	OpenAPI operation ids
Events	Event names/topics
Expected final state	Subscription active, WO finalized, invoice paid

11. Acceptance Checklist

- [ ] E2E test cases exist for all minimum scenarios.
- [ ] Each test references DDs and APIs.
- [ ] Test data seed is repeatable.
- [ ] Event consumers are tested for duplicate/replay.
- [ ] Frontend role smoke tests exist.
- [ ] Reporting reconciliation is tested.
- [ ] SIT/UAT entry and exit criteria are agreed.